

Optics Lens Assignment 1

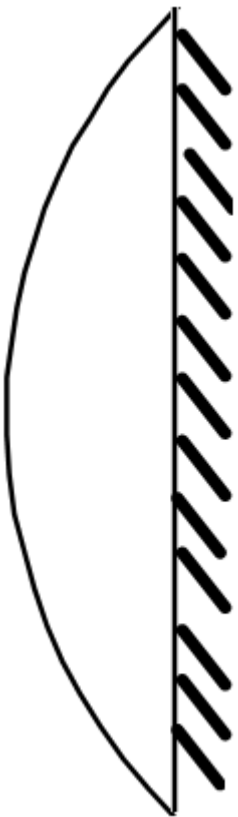
Physics

Section A

1. A plano-convex lens ($f = 20\text{ cm}$) is silvered at plane surface. Now focal length will be
A) 20 cm B) 40 cm C) 30 cm D) 10 cm

Answer: D,

Solution:



$$\frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_M}$$

For a plane mirror, $f_M = \infty$

Therefore $\frac{1}{F} = \frac{2}{f_L}$ or $F = \frac{f_L}{2} = \frac{20 \text{ cm}}{2} = 10 \text{ cm}$

2. The focal length of a convex lens of refractive index 1.5 is f when it is placed in air. when it is immersed in a liquid it behaves as a converging lens of focal length becomes xf ($x > 1$). The refractive index of the liquid
- A) Greater than $\frac{3}{2}$ B) Greater than 1 and Less than $\frac{3}{2}$ C) Less than $\frac{3}{2}$ D) All of these

Answer: B,

Solution:

$$1/f = (n - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{1.5}{1} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \text{ when the lens is plane in air and}$$

$$\frac{1}{xf} = \left(\frac{1.5}{y} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \text{ when the lens is plane in air and}$$

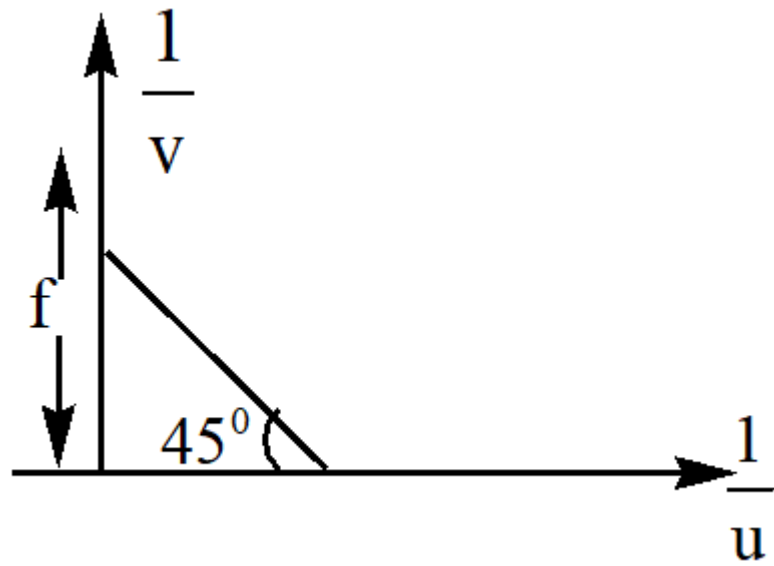
Where $y = R.I$ of the liquid.

On solving, we get

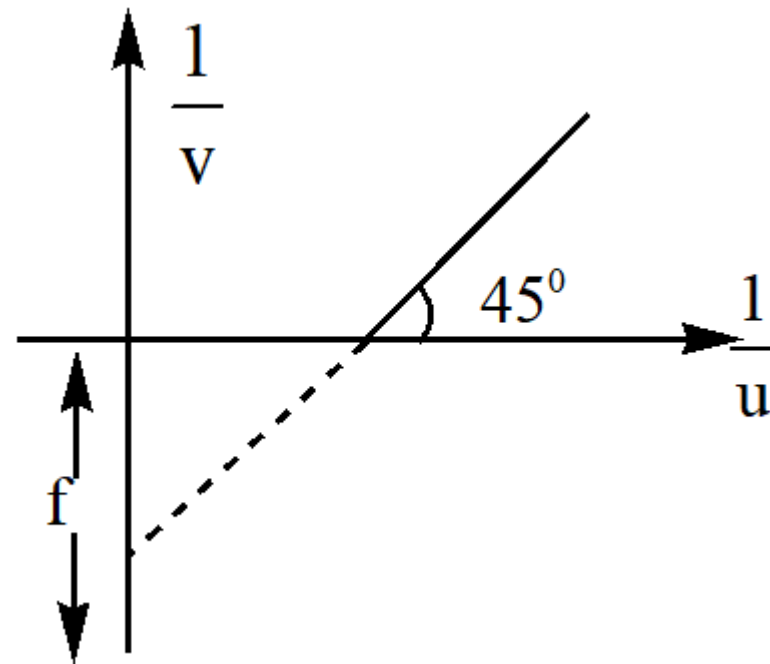
$$y = \frac{3}{2 + 1/x}.$$

3. For a given convex lens the $\frac{1}{u}, \frac{1}{v}$ graph for real object (u) and virtual image (v) is

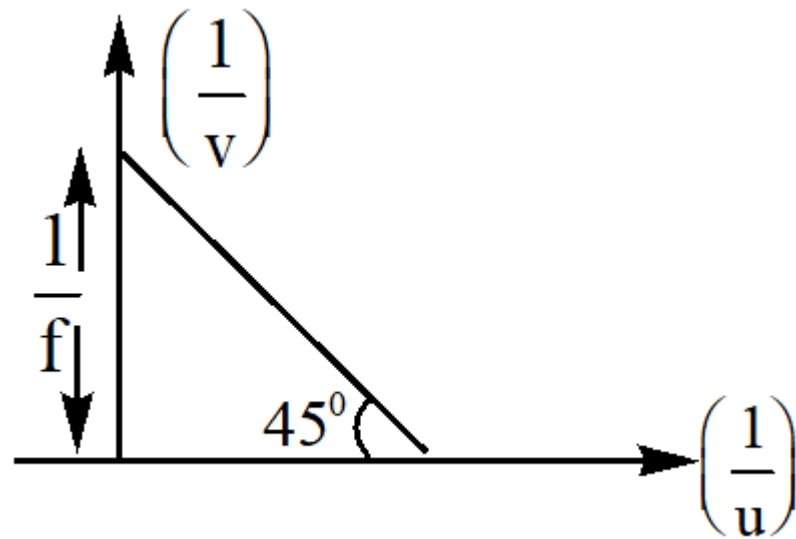
A)



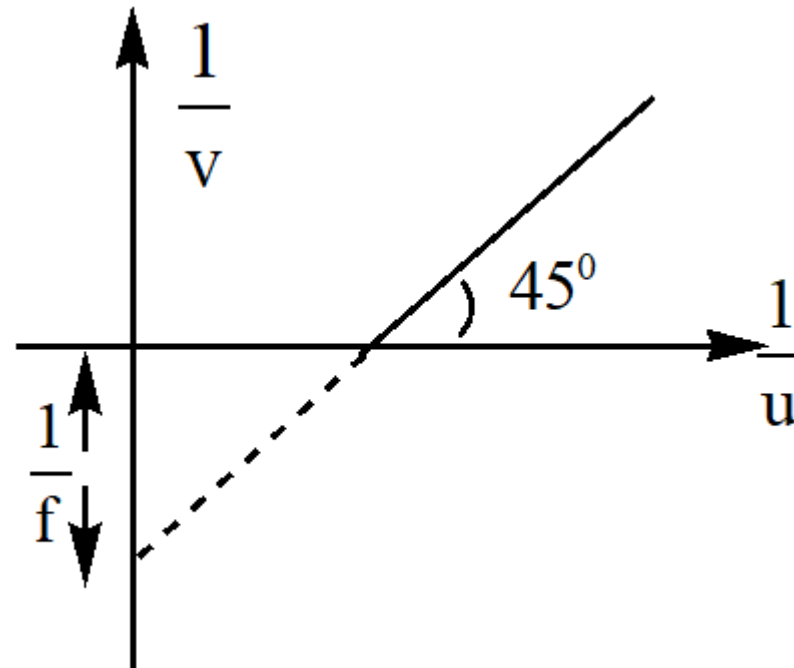
B)



C)



D)



Answer: D,

Solution:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = -\frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{u} - \frac{1}{f}$$

$$y = mx - c$$

+ve slope -ve intercept $\left(\frac{1}{f}\right)$ (45°)**4.** Arrange the following glass convex lenses in increasing order of their focal lengths_____

- a) $R_1 = 12 \text{ cm}, R_2 = 6 \text{ cm}$
 b) $R_1 = 10 \text{ cm}, R_2 = 5 \text{ cm}$
 c) $R_1 = 15 \text{ cm}, R_2 = 5 \text{ cm}$
 d) $R_1 = 15 \text{ cm}, R_2 = 10 \text{ cm}$
- A) a,b,c,d B) a,c,b,d C) b,c,a,d D) d,c,b,a

Answer: C,**Solution:**

From lens maker's formula, $\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

For convex lens, $R_1 = +ve$ and $R_2 = -ve$

$\therefore (a) R_1 = 12 \text{ cm}, R_2 = -6 \text{ cm}$

$\therefore \frac{1}{f_a} = c \left(\frac{1}{12} - \frac{1}{-6}\right)$ where c is constant.

$\frac{1}{f_a} = c \left(\frac{1}{12} + \frac{1}{6}\right) = \frac{c}{4}$

$\Rightarrow f_a = \frac{4}{c}$ -----(i)

(b) $R_1 = 10 \text{ cm}, R_2 = -5 \text{ cm}$

$\therefore \frac{1}{f_b} = c \left(\frac{1}{10} - \frac{1}{-5}\right) = c \left(\frac{1}{10} + \frac{1}{5}\right) = \frac{3}{10}c$

$\Rightarrow f_b = \frac{3.33}{c}$ ----- (ii)

(c) $R_1 = 15 \text{ cm}, R_2 = -5 \text{ cm}$

$\therefore \frac{1}{f_c} = c \left(\frac{1}{15} - \frac{1}{-5}\right) = \frac{4}{15}c$

$\Rightarrow f_c = \frac{15}{4c} = \frac{3.75}{c}$ ----- (iii)

(d) $R_1 = 15 \text{ cm}, R_2 = -10 \text{ cm}$

$\therefore \frac{1}{f_d} = c \left(\frac{1}{15} - \frac{1}{-10}\right) = \frac{25}{150}c = \frac{c}{6}$

$\Rightarrow f_d = \frac{6}{c}$ ----- (iv)

From equation (i), (ii), (iii) and (iv)

$f_b < f_c < f_a < f_d$

$\Rightarrow b, c, a, d$

5. An object is placed at 20cm from a convex mirror of focal length 10 cm. The image formed by a mirror is_____

- A) Real and at 20cm from the mirror B) Virtual and at 20cm from the mirror C) Virtual and at $20/3$ cm from the mirror D) Real and at $20/3$ cm from the mirror

Answer: C,

Solution:

$$f=10\text{cm}, u=-20\text{cm}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{-20} \Rightarrow \frac{1}{10} = \frac{1}{v} - \frac{1}{20} \Rightarrow$$

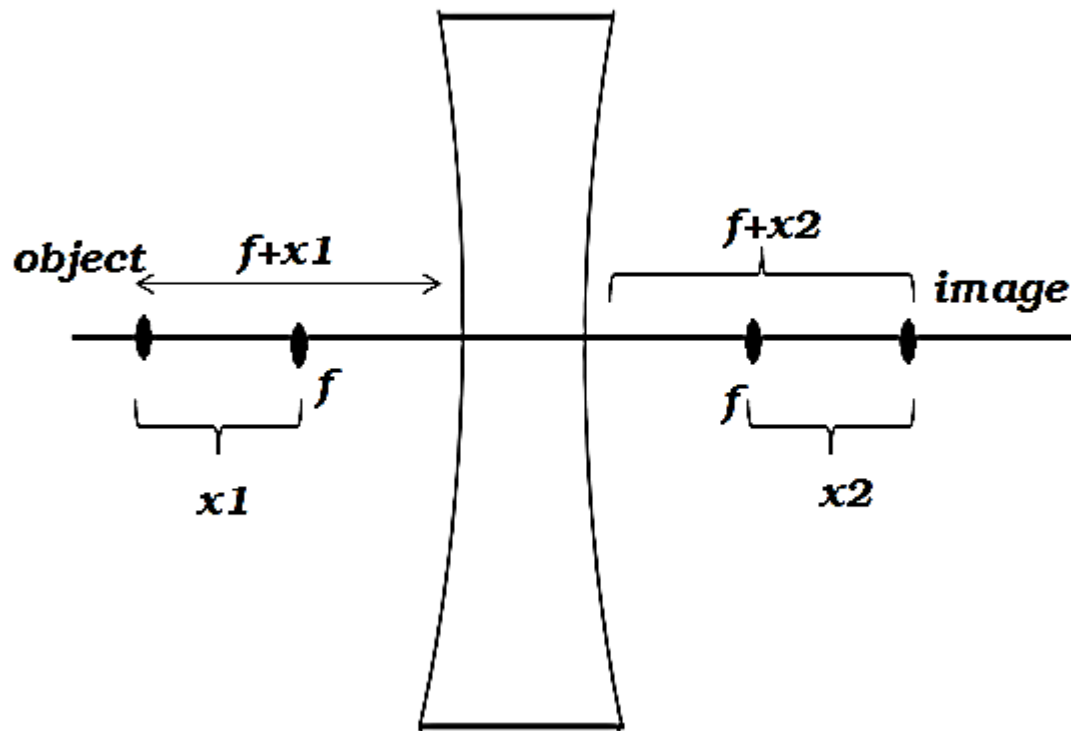
$$1/10+1/20=v=3/20\text{cm}$$

6. With a concave lens, an object is placed at a distance x_1 from the principal focus, on the principal axis. The image is formed at a distance x_2 from the principal focus. The focal length of the mirror is_____

- A) $x_1 x_2$ B) $\frac{X_1 + X_2}{2}$ C) $\sqrt{\frac{X_1}{X_2}}$ D) $\sqrt{X_1 X_2}$

Answer: D,

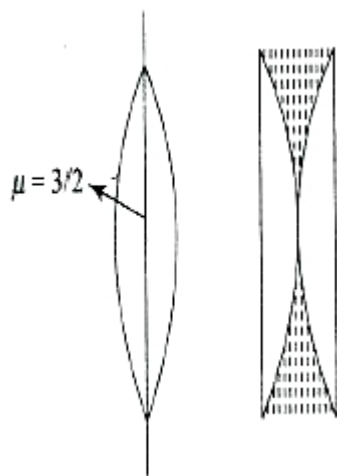
Solution:



Therefore $\frac{1}{f} = \frac{1}{f+x_2} - \frac{1}{-(f+x_1)}$

$$\begin{aligned} &= \frac{1}{f + x_2} + \frac{1}{f + x_1} = \frac{f + x_1 + f + x_2}{(f + x_2)(f + x_1)} \\ \text{Therefore } \frac{1}{f} &= \frac{2f + (x_1 + x_2)}{f^2 + fx_1 + fx_2 + x_1x_2} \\ 2f^2 + f(x_1 + x_2) &= f^2 + f(x_1 + x_2) + x_1x_2 \\ f^2 &= x_1x_2 \\ f &= \sqrt{x_1x_2} \end{aligned}$$

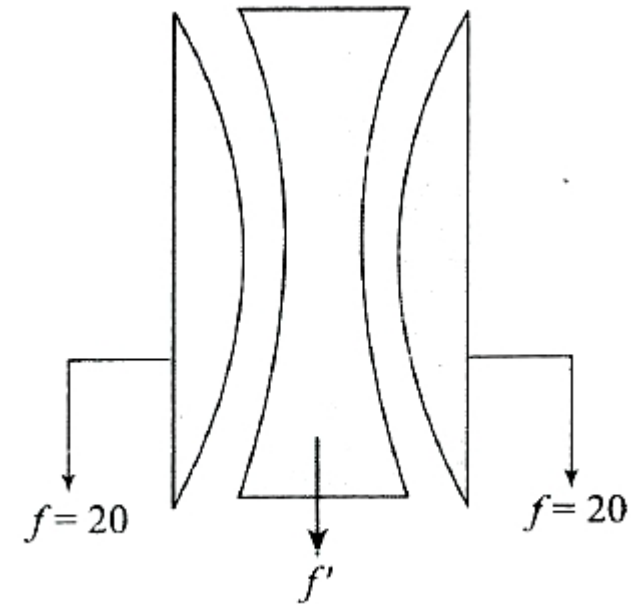
7. A thin equiconvex lens ($\mu = 3/2$) of focal length 10 cm is cut and separated and a material of refractive index 3 is filled between them. What is the focal length of the combination?



- A) -10cm B) -10/4cm C) -10/3cm D) None of these

Answer: C,
Solution:
$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

 $R = 10\text{cm}$



$$f' = (3 - 1) \left(\frac{1}{-10} - \frac{1}{10} \right)$$

$$f' = \frac{-10}{4}$$

$$\frac{1}{f_{eq}} = \frac{1}{20} - \frac{4}{10} + \frac{1}{20} = \frac{2}{20} - \frac{4}{10}$$

$$f_{eq} = -10/3 \text{ cm}$$

8. You are given two identical plano-convex lenses. When you place an object 20 cm to the left of a single plano-convex lens, the image appears 40 cm to the right of the lens. You then arrange the two plano-convex lenses back to back to form a double convex lens. If the object is 20 cm to the left of this new lens, what is the approximate location of the image?



Plano-convex

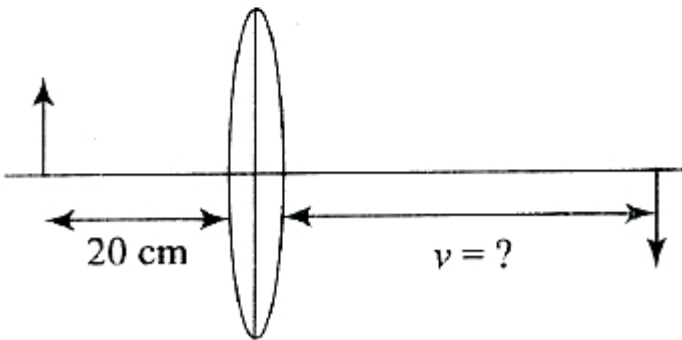
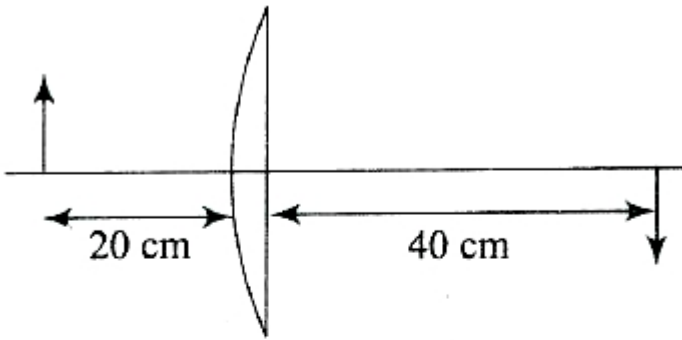


Double convex

- A) 10 cm to the right of the lens B) 20 cm to the right of the lens C) 80 cm to the right of the lens D) 80 cm to the left of the lens

Answer: A,

Solution:



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{40} - \frac{1}{(-20)}$$

$$\Rightarrow f = \frac{40}{3}$$

Focal length for the combination,

$$f_{eq} = \frac{20}{3}$$

$$\frac{1}{f_{eq}} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{\frac{20}{3}} = \frac{1}{v} + \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{3}{20}$$

$$v = 10\text{cm (to the right)}$$

9. The power of a lens having refractive index 1.25 is +3 diopters. When placed in a liquid its power is -2 diopters. The refractive index of the liquid is :

A) 1.2 B) 1.4 C) 1.5 D) 1.6

Answer: C,

Solution:

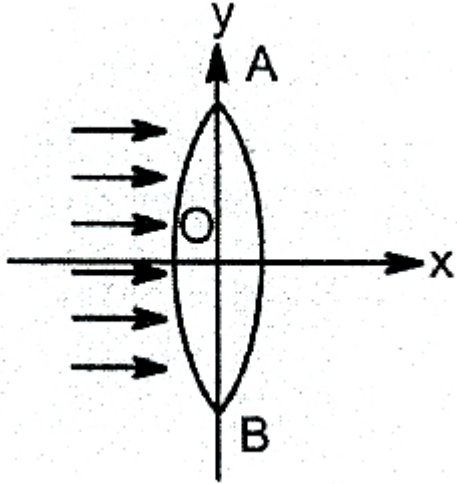
$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = 3$$

$$\therefore 3 = (1.25 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{ and } -2 = \left(\frac{1.25}{\mu} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$-\frac{3}{2} = \frac{0.25\mu}{1.25 - \mu} \Rightarrow -0.5\mu = 3.75 - 3\mu$$

$$\Rightarrow \mu = 3.75/2.5 = 1.5$$

10. Monochromatic light rays parallel to x -axis strike a convex lens AB . If the lens oscillates such that AB tilts upto a small angle θ (in radian) on either side of y -axis, then amplitude of oscillation of image will (f = focal length of the lens) :

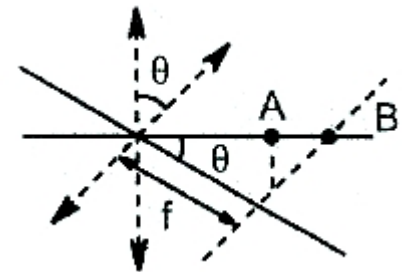


- A) $f \sec \theta$ B) $f \sec^2 \theta$ C) $\frac{f\theta^2}{2}$ D) $\frac{f\theta^2}{4}$

Answer: D,

Solution:

Along principal axis shift = $\frac{f}{\cos \theta} - f = f \left[\frac{1 - \cos \theta}{\cos \theta} \right] = \frac{f\theta^2}{2} \therefore \text{Amplitude} = \frac{f\theta^2}{4}$



11. A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of the combination is

- A) -6.5 D B) $+6.5 \text{ D}$ C) $+6.67 \text{ D}$ D) -1.5 D

Answer: D,

Solution:

$$P = P_1 + P_2 = \frac{1}{f_1 (m)} + \frac{1}{f_2 (m)} = \frac{1}{0.4} - \frac{1}{0.25} = -1.5D$$

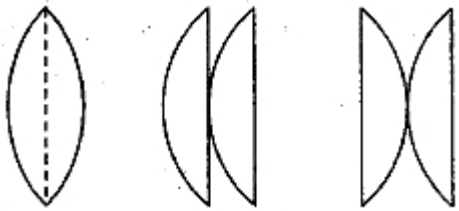
- 12.** Focal length of convex lens in air is $16cm$ ($\mu_{glass} = 1.5$). Now the lens is submerged in liquid of refractive index 1.42 . Find the ratio of focal length in medium to focal length in air has closest value
A) 9 B) 17 C) 1 D) 5

Answer: A,

Solution:

$$\begin{aligned} \frac{1}{f_a} &= \left(\frac{\mu_g}{\mu_a} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ \frac{1}{f_m} &= \left(\frac{\mu_g}{\mu_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ \Rightarrow \frac{f_a}{f_m} &= \frac{\left(\frac{\mu_g}{\mu_m} - 1 \right)}{\left(\frac{\mu_g}{\mu_a} - 1 \right)} = \frac{\left(\frac{1.50}{1.42} - 1 \right)}{\left[\frac{1.50}{1} - 1 \right]} = \frac{0.08}{(1.92)(0.5)} \\ \frac{f_m}{f_a} &= \frac{(1.42)(0.5)}{0.08} = 8.875 \approx 9 \end{aligned}$$

- 13.** Two similar plano-convex lenses are combined together in three different ways as shown in the figure. The ratio of focal lengths in three cases will be



- A) 2 : 1 : 1 B) 1 : 2 : 2 C) 2 : 2 : 1 D) 1 : 1 : 1

Answer: D,

Solution:

In every condition, two plano-convex lenses are placed, closed to each other i.e., the distance between is assumed to be zero.

i.e. $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

- 14.** A convex lens of focal length, f is placed somewhere in between an object and a screen. The distance between the object and the screen is x . If the numerical value of magnification produced by a lens is m , focal length of a lens is

- A) $\frac{mx}{(m+1)^2}$ B) $\frac{mx}{(m-1)^2}$ C) $\frac{(m+1)^2}{m}x$ D) $\frac{(m-1)^2}{m}x$

Answer: A,

Solution:

Here, $X = U + V$

As, $m = \frac{f}{f+u} = \frac{f-v}{f}$

and image is real, magnification is negative.

$$\therefore -m = \frac{f}{f+u}, u = \frac{-(m+1)f}{m}$$

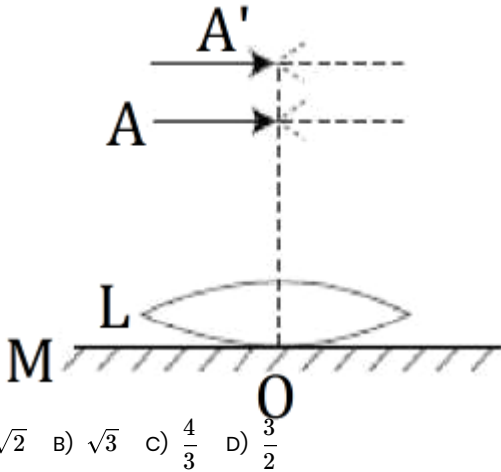
$$\text{From } -m = \frac{f-v}{f}$$

$$\Rightarrow v = (m+1)f$$

$$x = \frac{-(m+1)}{m}f + (m+1)f$$

$$\text{Solving, we get } f = \frac{mx}{(m+1)^2}$$

15. A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When pin is placed at A, such that $OA = 18 \text{ cm}$, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_l is put between the lens and the mirror, the pin has to be moved to A' , such that $OA' = 27 \text{ cm}$, to get its inverted real image at A' itself. The value of μ_l will be:



- A) $\sqrt{2}$ B) $\sqrt{3}$ C) $\frac{4}{3}$ D) $\frac{3}{2}$

Answer: C,

Solution:

At A

$$R_{\text{eff}} = 18 \text{ cm}$$

$$R_{\text{eff}} = -9 \text{ cm}$$

$$P_{\text{net}} = 2P_L + P_M$$

$$\frac{1}{9} = \frac{2}{f_L} + 0$$

$$\frac{1}{18}(\mu - 1) \left(\frac{2}{R_L} \right) \quad R_L = 18 \text{ cm}$$

At A'

$$R_{\text{eff}} = 27 \text{ cm}$$

$$R_{\text{eff}} = -27/2 \text{ cm}$$

$$P_{\text{net}} = 2P_{\text{cem}} + 2P_{\text{liquid}}$$

$$\frac{2}{27} = \frac{1}{9} + 2(\mu - 1) \left(\frac{-1}{18} \right)$$

$$\frac{1}{9} \left(\frac{-1}{3} \right) = 2(\mu - 1) \left(\frac{-1}{18} \right) \mu = \frac{4}{3}$$

